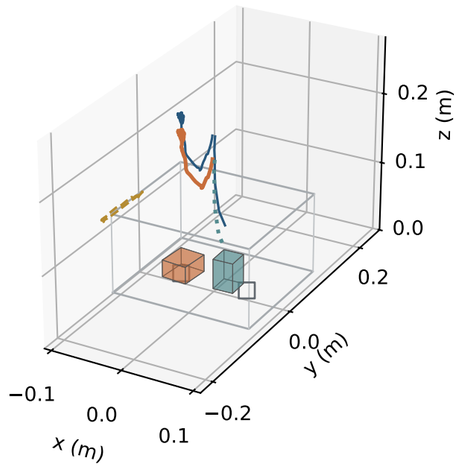
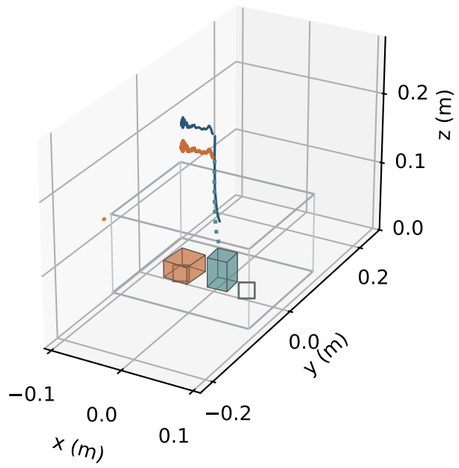
# Stage C.2 | Conformal risk-adaptive in-flow guidance

Left gripper · Right gripper · Item A · Item B

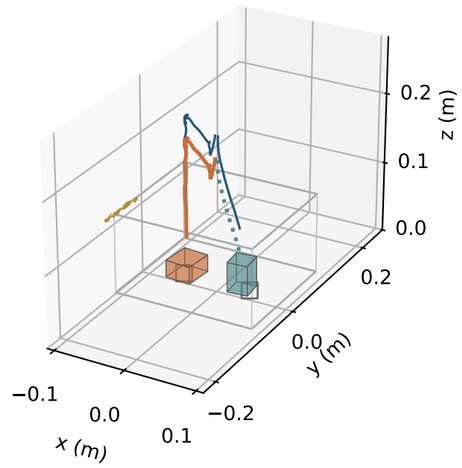
BC (H=4)
Packing: success | Collision: yes



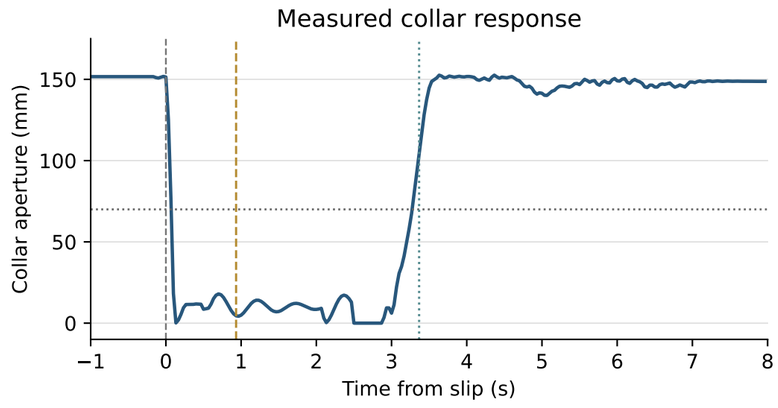

GPC-OPT (M=20)
Packing: success | Collision: no



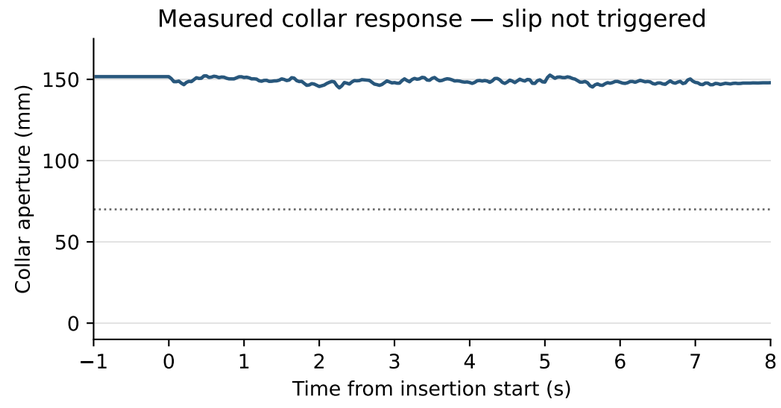

Ours (256D, adaptive tau <= 0.9)
Packing: success | Collision: no



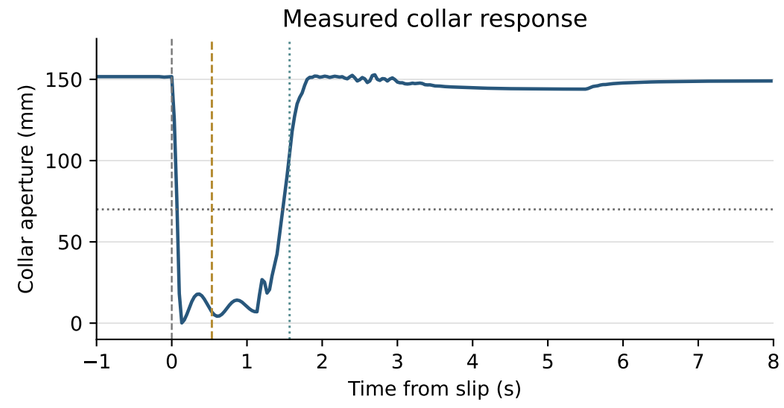

Measured collar response



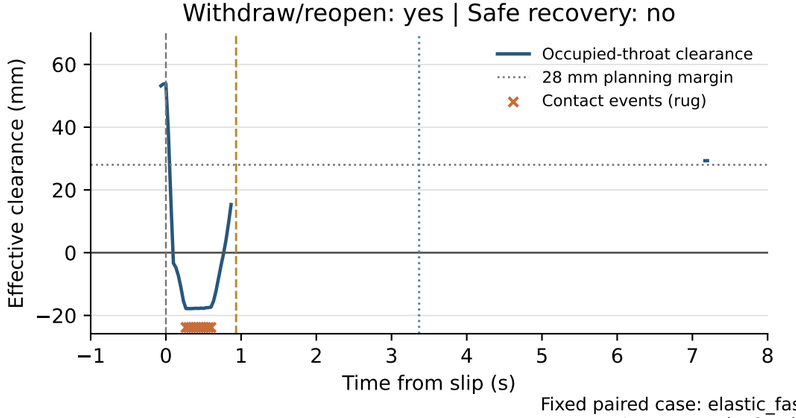

Measured collar response — slip not triggered



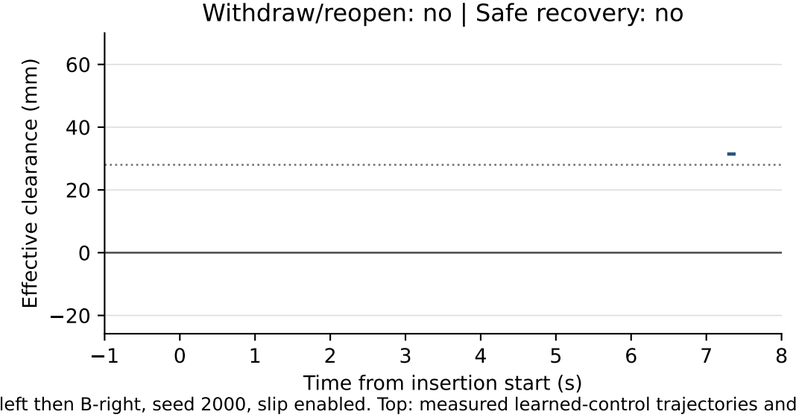

Measured collar response



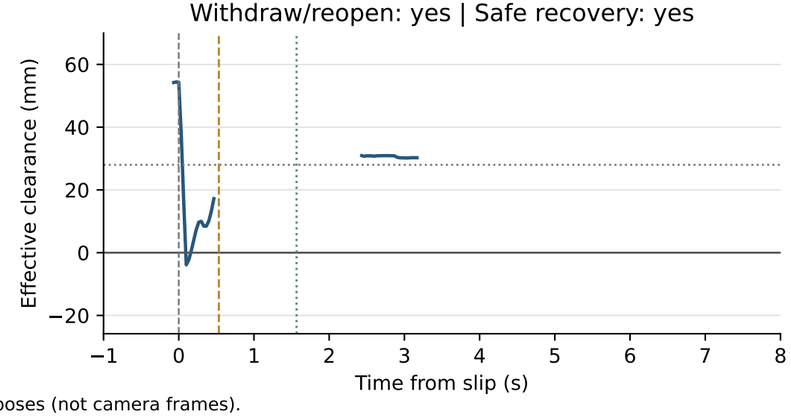

Withdraw/reopen: yes | Safe recovery: no




Withdraw/reopen: no | Safe recovery: no




Withdraw/reopen: yes | Safe recovery: yes




Fixed paired case: elastic_fast, A-left then B-right, seed 2000, slip enabled. Top: measured learned-control trajectories and final poses (not camera frames).
Lower panels: 9 s display windows; untriggered cases use insertion start. Clearance shown only while occupied; contact marks are a rug.
Dashed gold: measured withdrawal; dotted teal: measured reopening. Outcomes and unsuccessful recoveries are preserved.
Full-matrix safe recovery | BC H4: 0/18; GPC-OPT: 1/18; Ours tau=0.9: 18/18